

8. Astronomers have studied light from distant galaxies.
They found that the light from these galaxies has increased in wavelength.

(a) Complete each sentence below by **drawing a line** to the correct ending. [2]

(i)

The increase in wavelength of light
is due to the Universe...

contracting

staying the
same size

expanding

(ii)

The increase in wavelength of
light supports the Big Bang model
which states the Universe started
from...

different points

a single point

a supernova



(b) Look at the diagrams below.
They show absorption spectra of hydrogen in a laboratory on Earth (**Diagram A**), from a distant galaxy X (**Diagram B**) and from another distant galaxy Y (**Diagram C**).

Diagram A: Laboratory on Earth

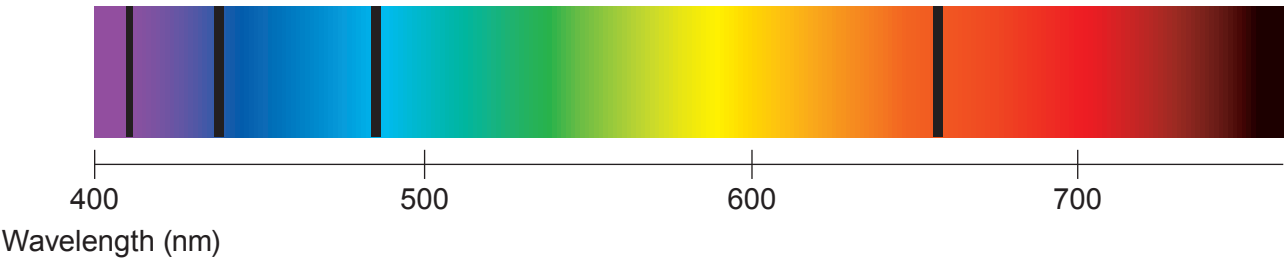


Diagram B: From galaxy X

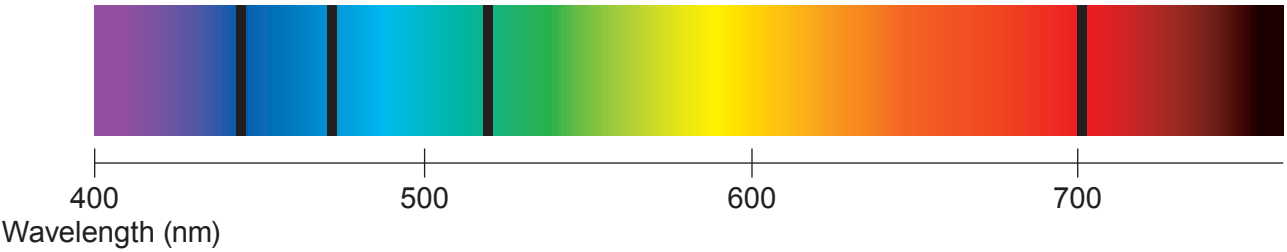
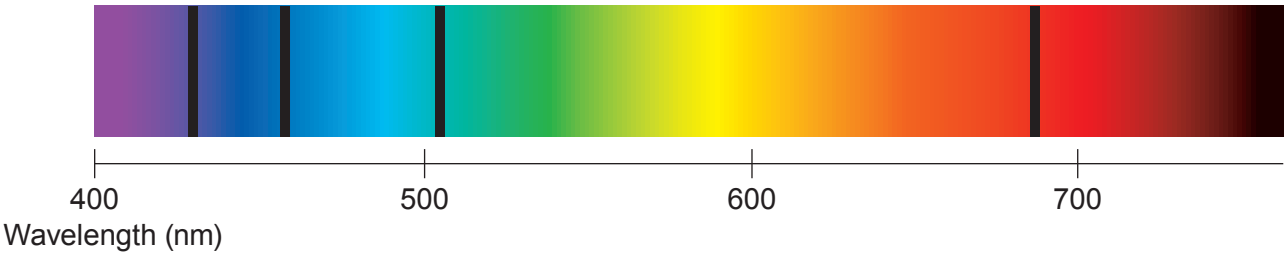


Diagram C: From galaxy Y



Examiner
only

Use the spectra opposite to answer the following questions.

(i) Compare the spectral dark lines in diagrams **A** and **B**. [2]

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(ii) State the **differences** between the speed **and** distance from Earth for galaxies **X** and **Y**. [2]

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